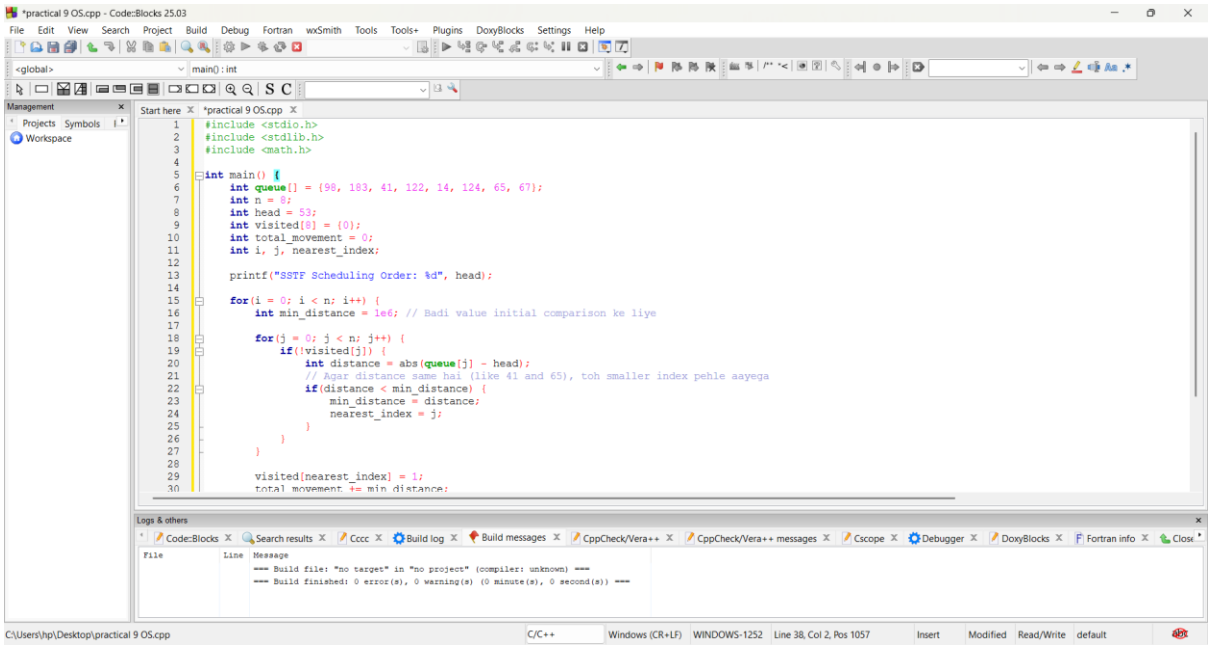
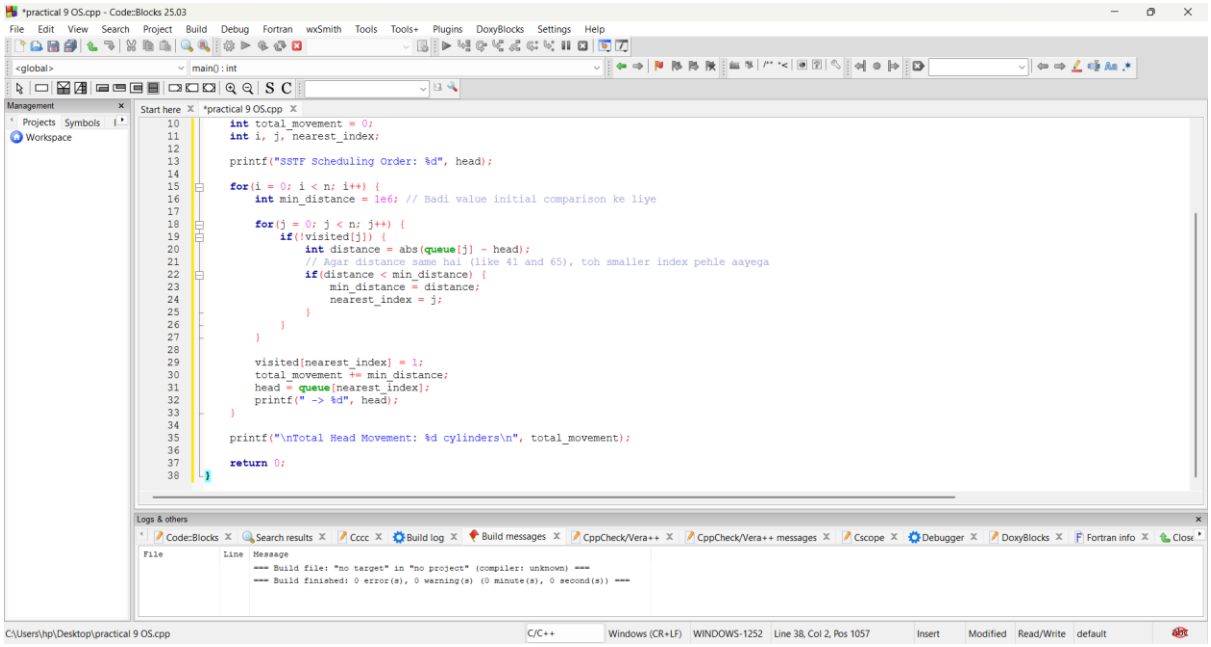
Aim: Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF (Shortest Seek Time First) scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a C Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.



```c
#include <stdio.h>
#include <stdlib.h>
#include <math.h>

int main() {
    int queue[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;
    int head = 53;
    int visited[8] = {0};
    int total_movement = 0;
    int i, j, nearest_index;

    printf("SSTF Scheduling Order: %d", head);

    for(i = 0; i < n; i++) {
        int min_distance = 1e6; // Badi value initial comparison ke liye

        for(j = 0; j < n; j++) {
            if(!visited[j]) {
                int distance = abs(queue[j] - head);
                // Agar distance same hai (like 41 and 65), toh smaller index pehle aayega
                if(distance < min_distance) {
                    min_distance = distance;
                    nearest_index = j;
                }
            }
        }

        visited[nearest_index] = 1;
        total_movement += min_distance;
```



```c
    int total_movement = 0;
    int i, j, nearest_index;

    printf("SSTF Scheduling Order: %d", head);

    for(i = 0; i < n; i++) {
        int min_distance = 1e6; // Badi value initial comparison ke liye

        for(j = 0; j < n; j++) {
            if(!visited[j]) {
                int distance = abs(queue[j] - head);
                // Agar distance same hai (like 41 and 65), toh smaller index pehle aayega
                if(distance < min_distance) {
                    min_distance = distance;
                    nearest_index = j;
                }
            }
        }

        visited[nearest_index] = 1;
        total_movement += min_distance;
        head = queue[nearest_index];
        printf(" -> %d", head);
    }

    printf("\nTotal Head Movement: %d cylinders\n", total_movement);

    return 0;
}
```

```
"C:\Users\hp\Desktop\practic

SSTF Scheduling Order: 53 -> 41 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 14
Total Head Movement: 323 cylinders

Process returned 0 (0x0)   execution time : 0.021 s
Press any key to continue.
```